

# Theoretic 0.3.1

## Contents

|                                                  |            |                              |     |
|--------------------------------------------------|------------|------------------------------|-----|
| <b>1 Summary</b>                                 | <b>1</b>   | Theorem A.3 Pythagoras       | A3  |
| Proposition 1.2 Foo                              | 1          | Theorem Only Named           | A3  |
| <b>2 Features</b>                                | <b>1</b>   | A.8 show-theorem . . . . .   | A4  |
| <b>3 Styling / Customization</b>                 | <b>3</b>   | A.9 solutions . . . . .      | A4  |
| Lemma 3.2 Title                                  | 3          | A.10 theorem . . . . .       | A4  |
| Lesson 3.6 Important Lesson                      | 3          | Theorem 2 Pythagoras         | A5  |
| 3.1 Preset Styles . . . . .                      | 4          | Theorem A.15 AAAAAA / ZZZZZZ | A7  |
| <b>A Detailed Documentation of the Functions</b> | <b>A1</b>  | Theorem A.17 Positional      | A7  |
| A.1 configure . . . . .                          | A1         | Theorem A.18 Named           | A7  |
| A.2 fill-options . . . . .                       | A1         | Theorem A.19 Mixed           | A7  |
| A.3 qed . . . . .                                | A1         | A.11 toc . . . . .           | A7  |
| A.4 qed-in-equation . . . . .                    | A2         | Theorem A.21 Z               | A9  |
| A.5 restate . . . . .                            | A2         | Theorem A.22 A               | A9  |
| Proposition A.1 Funky!                           | A2         | A.12 toc-entry . . . . .     | A9  |
| A.6 show-entry-as . . . . .                      | A3         | A.13 thm-counter . . . . .   | A11 |
| A.7 show-ref . . . . .                           | A3         | A.14 proof . . . . .         | A11 |
| <b>B Solutions</b>                               | <b>A11</b> |                              |     |

## 1 Summary

This package provides opinionated functions to create theorems and similar environments.

```
#import "@preview/theoretic:0.3.1"
#import theoretic.presets.basic: *
#show ref: theoretic.show-ref

#theorem[This is a theorem.]
#proof[
  This is a proof. A QED symbol is placed correctly even after
  block equations.
  $ norm(x) = sqrt( sum_(k = 1)^d x_k ) . $
]
#proposition(<thm:foo>)[Foo][This is a named theorem.]
#proof[@thm:foo[-]]
  Proof with a list or enum?
  - No problem for QED.
]
```

**Theorem 1.1.** *This is a theorem.*

*Proof.* This is a proof. A QED symbol is placed correctly even after block equations.

$$\|x\| = \sqrt{\sum_{k=1}^d x_k}.$$

**Proposition 1.2 (Foo).** *This is a named theorem.*

*Proof of Proposition 1.2.* Proof with a list or enum?

- No problem for QED.

## 2 Features

- Except for `#show ref: theoretic.show-ref`, no “setup” is strictly necessary.

Customisation of the environments is achieved via parameters on the `#theoretic.theorem()` function. You can use e.g. `#let lemma = theoretic.theorem.with(kind: "lemma", supplement: "Lemma", /* ... */)`. → See Section 3

For convenience, the `theoretic.presets` module contains predefined styled environments. → See Section 3.1

- Flexible References via specific supplements. → `#theoretic.show-ref()`

```
@thm:foo vs @thm:foo[-] vs @thm:foo[--] vs
@thm:foo[!] vs @thm:foo[!!] vs @thm:foo[!!!]
vs @thm:foo[?] vs @thm:foo[Statement]
```

Proposition 1.2 (Foo) vs Proposition 1.2 vs 1.2 vs Foo (Proposition 1.2) vs Foo (1.2) vs Foo vs Proposition vs Statement 1.2 (Foo)

- Any theorem can be restated. → `#theoretic.restate()`

```
theoretic.restate(<thm:foo>)
// the head links to the original
```

**Proposition 1.2** (Foo). *This is a named theorem.*

- Automatic numbering. If your headings are numbered, it will use top-level heading numbers as the first component, otherwise it will simply number your theorems starting with Theorem 1.

```
#theorem(number: "!!")[
  Number can be overridden for individual theorems.
]
#theorem(number: 40)[
  If a `number` is passed (as opposed to a string or content),
]
#theorem[
  ...subsequent theorems will pick it up.
]
```

**Theorem !!.** *Number can be overridden for individual theorems.*

**Theorem 2.40.** *If a number is passed (as opposed to a string or content),*

**Theorem 2.41.** *...subsequent theorems will pick it up.*

- Custom outlines: Outline for headings *and/or* theorems. → [#theoretic.toc\(\)](#)
  - Filter for specific kinds of theorem to create e.g. a list of definitions.
  - Optionally sorted alphabetically!
  - Theorems can have a different title for outlines ([theorem\(toctitle: ...\)](#)) and can even have multiple entries in a sorted outline.
  - Highly customizable! → [#theoretic.toc-entry\(\)](#)
    - (And this customization can be reused for regular outlines) → [#theoretic.show-entry-as\(\)](#)
- Automatic QED placement! → [#theorem\(suffix: ...\)](#) & [#theoretic.qed\(\)](#)

In most cases, it should place the QED symbol appropriately automatically:

```
#proof[This is a proof. $x=y$]
#proof[
  This is a proof.
  $ x = y $
]
#proof[
  #set math.equation(numbering: "(1)")
  This is a proof.
  $ x = y $
]
#proof(suffix: $smash$)[
  This is a proof.
  - #lorem(3)
]
#proof[
  This is a proof.
  - #lorem(3) $ x = y $
]
#proof[
  This is a proof.
  + #lorem(3)
  + #lorem(3)
  + #lorem(3)
  + #lorem(3)
]
]
```

*Proof.* This is a proof.  $x = y$  ■

*Proof.* This is a proof.  $x = y$  ■

*Proof.* This is a proof.  $x = y$  (1) ■

*Proof.* This is a proof. ■

► Lorem ipsum dolor. ✖

*Proof.* This is a proof.  $x = y$  ■

*Proof.* This is a proof. 1. Lorem ipsum dolor. 1. Lorem ipsum dolor. 1. Lorem ipsum dolor. 1. Lorem ipsum dolor. 1. Lorem ipsum dolor. ■

Specifically, it works for lists, enums, and unnumbered block equations, which may be nested. If your proof ends with some other block, you should might want to place a [#qed\(\)](#) manually.

If you need to place a qed manually which should be aligned to a block equation, put [#show: qed-in-equation](#) before said equation. → [#theoretic.qed-in-equation\(\)](#)

- Exercise solutions: → [#theoretic.solutions\(\)](#)
  - Every theorem environment can have a solution, which is shown in a separate section.
  - Solutions section automatically hides itself if there are no solutions to show.

```
#exercise(solution: [/*****/])[
  Go look for the solution of this exercise at the end of
  this document.
]
```

**Exercise 2.42.** Go look for the solution of this exercise at the end of this document.

### 3 Styling / Customization

For basic customization, you can override the supplement, kind, and options parameters of `#theoretic.theorem()`.

You can start completely fresh:

```
#import "@preview/theoretic:0.3.1"
#show ref: theoretic.show-ref

// simply setting `variant` to one of "plain", "remark", "definition" or "important" changes the style:
#let theorem = theoretic.theorem.with(supplement: "Theorem", kind: "theorem", variant: "important")
// you can also set the other style options directly:
#let lemma = theoretic.theorem.with(
  supplement: "Lemma", kind: "lemma",
  options: (
    head-font: (style: "normal", weight: "bold"),
    title-font: (style: "italic", weight: "regular", fill: oklch(40%, 0.2, 12deg)),
    body-font: (style: "normal", weight: "regular"),
    block-args: (outset: 4pt, fill: oklch(95%, 0.06, 12deg)),
    head-punct: none,
    head-sep: h(1em),
  ),
)

#lorem(5)
#theorem[#lorem(5)]
#lemma[Title][#lorem(5)]
```

Lorem ipsum dolor sit amet.

**Theorem 3.1.** Lorem ipsum dolor sit amet.

**Lemma 3.2 (Title)** Lorem ipsum dolor sit amet.

See the documentation in the appendix for more details on the options parameter.

Alternatively, you can also build upon a preset style:

```
#import "@preview/theoretic:0.3.1"
#import theoretic.presets.fancy: *
#show ref: theoretic.show-ref

// this is immediately useful for translations:
#let satz = theorem.with(supplement: "Satz")
#satz[Eine deutsche Aussage.]
#theorem[An English theorem.]

// or to add new kinds:
#let lesson = theorem.with(supplement: "Lesson", kind:
  "lesson", options: (color: yellow), variant: "plain")
#lesson[Bar]
#lesson(variant: "important")[Important Lesson][...]
```

**Satz 3.3** Eine deutsche Aussage.

**Theorem 3.4** An English theorem.

**Lesson 3.5** Bar

**Lesson 3.6 Important Lesson** The variant parameter is also intended to be called for single theorems. E.g. in this fancy preset, "important" adds a line above the supplement.

Note that not all preset styles respect the same options. More details are given in the examples in Section 3.1.

If you want to go in a completely new direction, you can also provide your own `show-theorem` function to fully control styling. For how this can look, I recommend looking at how the predefined styles are made: See the code on [GitHub](#).

### 3.1 Preset Styles

All preset styles define the following environments: `#algorithm[...]`, `#axiom[...]`, `#claim[...]`, `#corollary[...]`, `#counter-example[...]`, `#definition[...]`, `#example[...]`, `#exercise[...]`, `#lemma[...]`, `#note[...]`, `#proof[...]`, `#proposition[...]`, `#remark[...]`, and `#theorem[...]`.

#### Preset “basic”

Use with: `#import theoretic.presets.basic: *`

This style uses the built-in `#theoretic.show-theorem()` and accepts all its options.

This style supports the variants "definition", "plain", "important", "remark", and "proof"; and the options head-font, title-font, body-font, block-args, head-punct, head-sep, and link.

**Theorem 3.7** (Title). *This is an example theorem created using `#theorem(toctitle: none)[Title][...]`.*

*Proof.* Using `#proof[...]`. ■

*Proof of Title.* Using `#proof[...][...]`. ■

**Lemma 3.8.** Using `#lemma(variant: "important")[...]`.

**Definition 3.9.** Using `#definition[...]`.

*Remark.* Using `#remark[...]`.

**Example.** Using `#example[...]`.

#### Preset “corners”

Use with: `#import theoretic.presets.corners: *`

This style is almost identical to the basic one, it just wraps the environments in a block with corners. Note the added color option.

This style supports the variants "definition", "plain", "important", "remark", and "proof"; and the options head-font, title-font, body-font, block-args, head-punct, head-sep, link, and color.

**Theorem 3.10** (Title). *This is an example theorem created using `#theorem(toctitle: none)[Title][...]`.*

*Proof.* Using `#proof[...]`. ■

*Proof of Title.* Using `#proof[...][...]`. ■

**Lemma 3.11.** Using `#lemma(variant: "important")[...]`.

**Theorem 3.12** (Title). *Using `#theorem(options: (color: red), ..)`.*

**Definition 3.13.** Using `#definition[...]`.

*Remark.* Using `#remark[...]`.

**Example.** Using `#example[...]`.

#### Preset “fancy”

Use with: `#import theoretic.presets.fancy: *`

This style is intended<sup>1</sup> for use with a font that supports stretch: 85% and weight: "semibold". It is here shown using the *Besley\** font, which you can download on GitHub.

This style supports the variants "plain", "important", "muted", and "remark"; and the options color, head-font, body-font, block-args, and link.

**Theorem 3.15** **Title** This is an example theorem created using `#theorem(toctitle: none)[Title][...]`.

*Proof.* Using `#proof[...]`. □

*Proof of Title.* Using `#proof[...][...]`. □

**Lemma 3.16** Using `#lemma(variant: "plain")[...]`.

**Lemma 3.17** Using `#lemma(variant: "important")[...]`.

**Theorem 3.18** **Title** Using `#theorem(options: (color: red), ..)`.

**Definition 3.19** Using `#definition[...]`.

*Remark* Using `#remark[...]`.

Example 3.20 Using `#example[...]`.

<sup>1</sup>It still looks okay in other fonts, but it does not reach full potential. Compare: **Lemma 3.14** **Lorem** ipsum.

### Preset “bar”

This style supports the variants "plain", and "important"; and the options head-font, link, and color.

#### Theorem 3.21

#### Title

This is an example theorem created using `#theorem(toctitle: none)[Title][...]`.

*Proof:* Using `#proof[...]`. ■

*Proof of Title:* Using `#proof[...][...]`. ■

#### Lemma 3.22

Using `#lemma(variant: "important")[...]`.

Use with: `#import theoretic.presets.bar: *`

#### Theorem 3.23

#### Title

Using `#theorem(options: (color: red), ..)`.

#### Definition 3.24

Using `#definition[...]`.

#### Remark

Using `#remark[...]`.

#### Example

Using `#example[...]`.

### Preset “columns”

This style supports the variants "plain", "important", "remark", and "proof"; and the options head-font, number-font, title-font, body-font, block-args, grid-args, head-punct, and link.

**Theorem**      This is an example theorem created using  
3.25              `#theorem(toctitle: none)[Title][...]`  
*Title*

*Proof.* Using `#proof[...]`. ■

Use with: `#import theoretic.presets.columns: *`

*Proof of Title.* Using `#proof[...][...]`. ■

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| <b>Lemma</b><br>3.26 | Using <code>#lemma(variant: "important")[...]</code> . |
|----------------------|--------------------------------------------------------|

|                           |                                       |
|---------------------------|---------------------------------------|
| <b>Definition</b><br>3.27 | Using <code>#definition[...]</code> . |
|---------------------------|---------------------------------------|

|               |                                   |
|---------------|-----------------------------------|
| <b>Remark</b> | Using <code>#remark[...]</code> . |
|---------------|-----------------------------------|

|                |                                    |
|----------------|------------------------------------|
| <b>Example</b> | Using <code>#example[...]</code> . |
|----------------|------------------------------------|

### Preset “colorbox”

Use with: `#import theoretic.presets.colorbox: *`

This is a more flashy style. You may want to combine it with another preset to avoid emphasizing too many things, e.g. by using `#example[]` from the basic preset.

This style supports the variants "plain", "remark", and "proof"; and the options head-font, title-font, body-font, block-args, head-punct, color, icon, and link.

#### Theorem 3.28 (Title)

This is an example theorem created using `#theorem(toctitle: none)[Title][...]`. ♥

*Proof.* Using `#proof[...]`. ■

*Proof of Title.* Using `#proof[...][...]`. ■

#### Theorem 3.29 (Title)

Using `#theorem(options: (color: red), ..)`. ♥

#### Definition 3.30

Using `#definition[...]`. ♣

#### Remark

Using `#remark[...]`. ♦

#### Example

Using `#example[...]`. ♦

## A Detailed Documentation of the Functions

### A.1 configure

Allows overriding some of the `theorem()` parameters.

All parameters correspond to the parameter of the same name of `theorem()`. Setting it here redefines what `auto` means for that parameter.

You might want to update `thm-counter` to 0 after calling `configure()`.

#### Parameters

`configure`(numbering: `str` `function`, numbering-depth: `int`)

### A.2 fill-options

Used to fill the `show-theorem()` `it.options` with default values.

If you create your own `show-theorem` function, you should make sure to use this or something similar to handle unset options.

This function is intended for use *only* when creating your own style, have a look at the `columns` style as to how this can be used.

Note: the output here depends on the chosen variant. Variants `plain`, `definition`, `remark` correspond to the respective `amsthm` styles if combined with the default `show-theorem()`. Variant `important` is like `definition`, but with an added border.

#### Parameters

`fill-options`(<options>: `dictionary`, variant: `str`, \_defaults: `dictionary`)

\_defaults: `dictionary`

A dictionary containing the default values for each variant. If the variant passed is `proof`, it will fill using the last entry in this dictionary, otherwise if the variant passed is not a key of the dictionary, it will use the first one.

### A.3 qed

Place the QED mark last pushed to `state("_thm_qed_stack")` here.

Will be automatically called by theorems to place their suffix. You should only need to call this in the rare situation that theoremic cannot place the qed in the correct position automatically. (If this occurs, please let me know on GitHub or Typst Forum so I can try to fix it.)

```
#proof[
  I want the QED on this line already.#qed()

  There will be no QED on this line, even though this is still
  in the proof.
]
```

*Proof.* I want the QED on this line already. ■  
There will be no QED on this line, even though this is still in the proof.

#### Parameters

`qed`(<symbol>: `content`) → `content`

<symbol> `content` Positional

Optional.

If provided, will ignore the `state("_thm_qed_stack")` and just place the given symbol here. Use this if you want to call it outside of a `proof` or other theorem with suffix.

This is not inside a proof environment.`#qed($triangle$)`

This is not inside a proof environment.  $\triangle$

## A.4 qed-in-equation

Use this to place a QED in the (next) block equation. Can be called either as with `#show:` or simply by wrapping the block equation.

```
#import theoretic: qed-in-equation
#proof[
  #show: qed-in-equation
  $ x = y $
  QED is above this line!
]
#proof[
  #qed-in-equation($ x = y $)
  QED is above this line!
]
```

*Proof.*

$$x = y$$

■

QED is above this line!

*Proof.*

$$x = y$$

■

QED is above this line!

### Parameters

`qed-in-equation(<rest>)`

## A.5 restate

Re-state a theorem.

It will reuse the original kind, supplement, number, title, body, and styling. It will *not* re-emit the solution or label, and it will use `toctitle: none` to avoid duplicate toc entries.

```
#let proposition = theorem.with(
  kind: "proposition",
  supplement: "Proposition",
  options: (
    head-font: (fill: blue),
  )
)
#proposition(<funky>)[Funky!][Blah _blah_ blah.]
Restated:
#restate("funky")
Restated with added customizations:
#restate(<funky>, options: (body-font: (fill: red)))
```

**Proposition A.1 (Funky!).** *Blah blah blah.*

Restated:

**Proposition A.1 (Funky!).** *Blah blah blah.*

Restated with added customizations:

**Proposition A.1 (Funky!).** *Blah blah blah.*

### Parameters

`restate(<label>: label string, <..args>) → content`

`<label>`    `label` or `string`    Positional    Label of the theorem to restate.

`<..args>`    Positional    Override arguments for the theorem function.

(I don't recommend changing anything here except possibly `show-theorem` and `options`.)

## A.6 show-entry-as

Helper function to adapt actual outlines to look the same as those made with `toc()`. This is useful if you want to have e.g. a list of figures and a list of definitions and want them to share their style.

Note: For `typst` versions  $\leq 0.12$ , this function is a bit “hacky” and might not always work. (It deconstructs the `outline.entry` based on heuristics.)

```
#import theoretic: show-entry-as, toc-entry

#outline(target: figure, title: [Typst Default])

#show outline.entry: show-entry-as(toc-entry.with(hanging-indent: 60pt, /*...*/))
#outline(target: figure, title: [Using `theoretic.toc-entry`])

#figure(
  caption: [Example Figure],
  block(height: 2em, width: 100%, fill:
    gradient.linear(..color.map.viridis))
)
```

### Typst Default

Figure 1 Example Figure ..... A3

### Using `theoretic.toc-entry`

Figure 1 Example Figure ..... A3



Figure 1: Example Figure

### Parameters

`show-entry-as`(`<toc-entry>`): `function`)

`<toc-entry>` `function` Positional

Customize `toc-entry()` used.

Expects a function taking five positional arguments (level, target, prefix, body, page).

## A.7 show-ref

Show-rule-function to be able to @ labelled theorems.

Use via `#show ref: show-ref` at the beginning of your document.

```
#show ref: theoretic.show-ref
#theorem(label: <fact>, supplement: "Fact")[#lorem(2)]
#theorem(<pythagoras>, "Pythagoras")[#lorem(2)]
#theorem(label: <z1>, title: "Only Named", number: none)
[#lorem(2)]
#theorem(label: <y>, number: "Y")[#lorem(2)]
#theorem(label: "5", number: none)[#lorem(2)]
```

As a consequence of `@fact` and `@pythagoras[!]`...

**Fact A.2.** *Lorem ipsum.*

**Theorem A.3** (Pythagoras). *Lorem ipsum.*

**Theorem** (Only Named). *Lorem ipsum.*

**Theorem Y.** *Lorem ipsum.*

**Theorem.** *Lorem ipsum.*

As a consequence of Fact A.2 and Pythagoras (A.3)...

The reference can be controlled via the supplement passed:

|                                          | BOTH                     | WITHOUT TITLE        | WITHOUT NUMBER       | NEITHER |
|------------------------------------------|--------------------------|----------------------|----------------------|---------|
| <code>@ref</code> (Full)                 | Theorem A.3 (Pythagoras) | Fact A.2 / Theorem Y | Theorem (Only Named) | Theorem |
| <code>@ref[-]</code> (Compact)           | Theorem A.3              | Fact A.2 / Theorem Y | Theorem (Only Named) | Theorem |
| <code>@ref[--]</code> (Number)           | A.3                      | A.2 / Y              | (Only Named)         | Theorem |
| <code>@ref[!]</code> (Inverted)          | Pythagoras (Theorem A.3) | Fact A.2 / Theorem Y | Only Named (Theorem) | Theorem |
| <code>@ref[!!]</code> (Compact Inverted) | Pythagoras (A.3)         | Fact A.2 / Theorem Y | Only Named           | Theorem |
| <code>@ref[!!!]</code> (Name)            | Pythagoras               | Fact A.2 / Theorem Y | Only Named           | Theorem |

|                                               |                                |                              |                            |                |
|-----------------------------------------------|--------------------------------|------------------------------|----------------------------|----------------|
| <code>@ref[?]</code> (Kind)                   | <u>Theorem</u>                 | <u>Fact / Theorem</u>        | <u>Theorem</u>             | <u>Theorem</u> |
| <code>@ref[Custom]</code> (Custom Supplement) | <u>Custom A.3 (Pythagoras)</u> | <u>Custom A.2 / Custom Y</u> | <u>Custom (Only Named)</u> | <u>Custom</u>  |

Note: the fact that references and links in this document are underlined in gray is achieved with a separate `@show` link: `it ⇒ underline(..)` rule, and not because of this function.

### Parameters

`show-ref(<it>: ref)`

## A.8 show-theorem

Default “show” function for theorems. Note that in your versions of this, you cannot use it to generate the default options, but you can fall back to `theoretic.show.theorem(it)`.

For your own style, make sure to always handle the “link” option, which will be set by `restate()` and `solutions()` and contains a link target for the supplement (to link to the original location).

### Parameters

`show-theorem(<it>: dictionary)`

`<it>` `dictionary` Positional

A dictionary with keys:

- `supplement`: `content`
- `number`: `content` | `none`
- `title`: `content` | `none`
- `body`: `content`
- `variant`: `str`
- `options`: `dictionary`

Note that the suffix (QED) is already added to the body at this point.

Also, note that the variant is already used when filling the options dictionary with defaults. For the expected keys of options, see `theorem.options`

## A.9 solutions

List all solutions, if any. Should not be used more than once in a document, as this might break links.

See Section B for how it looks.

### Parameters

`solutions(title: content, theorem-function: function, supplement: content) → content`

`title`: `"Solutions"` `content` Title/heading to use.

`theorem-function`: `theorem` `function` Which theorem function to use for the solutions.

`supplement`: `"Solution"` `content` Supplement to use for the solution theorems. Change this e.g. for localization.

## A.10 theorem

Theorem Environment

```
#set heading(numbering: none)

#theorem[If the headings are not numbered, theorem numbering
starts at 1.]

= Heading
#theorem(title: "Pythagoras")[
  Given a right-angled triangle, the length
  of the hypotenuse squared is equal to the
  sum of the squares of the remaining sides'
  lengths.
]
```

**Theorem 1.** *If the headings are not numbered, theorem numbering starts at 1.*

## Heading

**Theorem 2** (Pythagoras). *Given a right-angled triangle, the length of the hypotenuse squared is equal to the sum of the squares of the remaining sides' lengths.*

## Parameters

```
theorem(
  show-theorem: function,
  options: dictionary,
  variant: str,
  suffix: none content,
  kind: string,
  supplement: content,
  number: auto none int content,
  numbering: auto str function,
  numbering-depth: auto int,
  title: none content,
  toctitle: auto none content array,
  label: none label string,
  solution: none content,
  <..unnamed-and-body>: arguments
) → content
```

**show-theorem:** show-theorem    **function**

This function is used to show the actual theorem. I recommend looking at the code and documentation for the default `show-theorem()` to see how this would look.

**options:** (:)    **dictionary**    Additional options that are passed to show-theorem.

The default show-theorem handles the following keys:

- head-font: dict    // options for the head text
- title-font: dict    // title is placed inside the head
- body-font: dict    // options for the body text
- block-args: dict    // options for the block
- head-punct: content // placed at the end of the head
- head-sep: content    // placed after the head
- link: none | (link target) // The target the head should link to.

If you are using a custom show-theorem, you can also add other fields here.

This will be filled with defaults depending on the variant.

**variant:** "plain"    **str**    This controls the defaults for options. It is also passed to show-theorem.

**suffix:** none    **none** or **content**    Will be placed at the end of the theorem or where `qed()` is called.

```
#theorem(suffix: sym.suit.spade)[...]
#proof(
  suffix: $limits(script(square))^arrow.zigzag$,
)[Proof by contradiction!]
```

**Theorem A.3.** ...

*Proof.* Proof by contradiction!



**kind:** "theorem" string Used for filtering e.g. when creating table of theorems.

**supplement:** "Theorem" content

What to label the environment.

It is recommended to keep kind and supplement matching (except for “subtypes”, e.g. one might have the kind of “Example” and “Counter-Example” both as "example")

**number:** auto auto or none or int or content

- If **auto**, will continue numbering from last numbered theorem.
- If **int**, it will continue the numbering of later theorems from the given number.
- If **content**, it is shown as-is, with no side-effects.

```
#let corollary = theorem.with(  
  kind: "corollary",  
  supplement: "Corollary")  
  
#corollary[#lorem(2)]  
  
#corollary(number: none)[Skip number]  
#corollary[Resume numbering]  
  
#corollary(number: "X")[Custom "number"]  
#corollary[Resume numbering]  
  
#corollary(number: 10)[Set number]  
#corollary[Continue from set number]
```

**Corollary A.4.** Lorem ipsum.

**Corollary.** Skip number

**Corollary A.5.** Resume numbering

**Corollary X.** Custom “number”

**Corollary A.6.** Resume numbering

**Corollary A.10.** Set number

**Corollary A.11.** Continue from set number

**numbering:** auto auto or str or function

Numbering to use for the theorem number itself. (May be preceded by the heading number)

Can be set for all theorems using `configure()`.

**numbering-depth:** auto auto or int

How many levels of the heading counter the theorem numbering should inherit. (Ignored if headings are not numbered)

Can be set for all theorems using `configure()`.

Note that this uses the numbering set on `heading()`, which may differ from the numbering for `heading(level: ..)`. Make sure to `#set heading(numbering: ..)` for the intended result. Note that changing the numbering-depth does not automatically reset the number.

```
#corollary(numbering-depth: 0)[blah]  
#corollary(numbering-depth: 1)[blah]  
#corollary(numbering-depth: 2)[blah]
```

**Corollary 12.** blah

**Corollary A.13.** blah

**Corollary A.10.14.** blah

**title:** none none or content Title of the Theorem. Usually shown in parentheses after the number.

*This can also be passed as a positional argument.*

**toctitle:** auto auto or none or content or array

Title of the Theorem to be used in outlines.

- **auto** to use the title.
- **none** to hide it from the outlines.

If you pass an array, in *sorted* outlines (`toc.sort`) it will be split into multiple entries. All but the first one are marked as secondary.

```
#theorem(  
  title: [A to Z],  
  toctitle: ([AAAAA], [ZZZZZZ])  
)[  
  Compare how this appears in different outlines!  
]
```

**Theorem A.15** (A to Z). *Compare how this appears in different outlines!*

**label:** `none` or `label` or `string`

Label (for references)

*This can also be passed as a positional argument. In that case it must be a label and not a string.*

NB: Simply putting a `<label>` after the `#theorem[]` does not work for referencing.

**solution:** `none` or `content`

Optional Solution. See also `solutions()`.

```
#theorem(solution: [This will show up wherever  
`#theoretic.solutions()` is placed.])[#lorem(5)]
```

**Theorem A.16.** *Lorem ipsum dolor sit amet.*

`<..unnamed-and-body>` arguments Positional

The last positional argument given is used as the theorem body.

Other positional arguments are used for the title and label, depending on their type.

```
// Any of these work:  
#theorem(<positional>)[Positional][#lorem(4)]  
#theorem(label: <named>, title: [Named])[#lorem(4)]  
#theorem([Mixed], label: <mixed>)[#lorem(4)]  
#theorem(<untitled>)[#lorem(4)]
```

**Theorem A.17** (Positional). *Lorem ipsum dolor sit.*

**Theorem A.18** (Named). *Lorem ipsum dolor sit.*

**Theorem A.19** (Mixed). *Lorem ipsum dolor sit.*

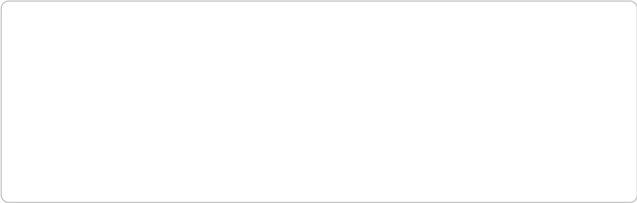
**Theorem A.20.** *Lorem ipsum dolor sit.*

## A.11 toc

Create an outline that includes named theorems.

Can be styled with show rules for `outline.entry()`. See the source code of this manual for an example.

```
#heading(outlined: false, level: 3)[  
  Contents  
]  
#toc(depth: 1)
```



**Contents**

1 Summary ..... 1  
    Proposition 1.2 Foo ..... 1  
2 Features ..... 1  
3 Styling / Customization ..... 3  
    Lemma 3.2 Title ..... 3  
    Lesson 3.6 Important Lesson ..... 3  
A Detailed Documentation of the Functions ..... A1  
    Proposition A.1 Funky! ..... A2  
    Theorem A.3 Pythagoras ..... A3  
    Theorem Only Named ..... A3  
    Theorem 2 Pythagoras ..... A5  
    Theorem A.15 AAAAAA / ZZZZZZ ..... A7  
    Theorem A.17 Positional ..... A7  
    Theorem A.18 Named ..... A7  
    Theorem A.19 Mixed ..... A7  
    Theorem A.21 Z ..... A9  
    Theorem A.22 A ..... A9  
B Solutions ..... A11

**Parameters**

`toc(depth: int, exclude: array (string), level: int auto, toc-entry: function, sort: bool) → content`

**depth:** 2 int Maximum depth of headings to consider

**exclude:** ("proof", "solution") array (string) list of theorem.kinds to ignore.

```
#heading(outlined: false, level: 3)[
  Table of Examples
]
#toc(
  depth: 0,
  exclude: ("proof", "solution", "theorem")
)
```

**Table of Examples**

Proposition 1.2 Foo ..... 1  
Lemma 3.2 Title ..... 3  
Lesson 3.6 Important Lesson ..... 3  
Proposition A.1 Funky! ..... A2

**level:** auto int or auto Fake level to use for theorems. If auto, it will use depth + 1.

**toc-entry:** toc-entry function

Customize toc-entry() used.

Expects a function taking five positional arguments (level, target, prefix, body, page).

**sort:** false bool Whether to sort the entries alphabetically.

Only respected if depth is 0.

If true, this will also split entries where toctitle is an array into separate entries.

```
#theorem("Z")[Blah blah.]
#theorem("A")[Blah blah.]
#heading(outlined: false, level: 3)[
  Sorted Table of Theorems
]
#set text(size: 9pt)
#toc(
  depth: 0,
  sort: true,
```

```
toc-entry: toc-entry.with(hanging-indent: 60pt),
)
```

**Theorem A.21 (Z).** *Blah blah.*

**Theorem A.22 (A).** *Blah blah.*

#### Sorted Table of Theorems

|                 |                        |    |
|-----------------|------------------------|----|
| Theorem A.22    | A .....                | A9 |
| Theorem A.15    | AAAAA .....            | A7 |
| Proposition 1.2 | Foo .....              | 1  |
| Proposition A.1 | Funky! .....           | A2 |
| Lesson 3.6      | Important Lesson ..... | 3  |
| Theorem A.19    | Mixed .....            | A7 |
| Theorem A.18    | Named .....            | A7 |
| Theorem         | Only Named .....       | A3 |
| Theorem A.17    | Positional .....       | A7 |
| Theorem A.3     | Pythagoras .....       | A3 |
| Theorem 2       | Pythagoras .....       | A5 |
| Lemma 3.2       | Title .....            | 3  |
| Theorem A.21    | Z .....                | A9 |
| Theorem A.15    | (ZZZZZZ) .....         | A7 |

## A.12 toc-entry

Create a toc entry.

Pass this to `toc()` using `.with(..)` to customize the `fmt-` parameters used.

This is used because since Typst 0.13, it is no longer possible to call `outline.entry` outside of an actual outline element, and one “cannot outline metadata”.

This manual uses

```
set par(justify: false)
let indents = (0pt, 15pt, 37pt)
let hang-indents = (15pt, 22pt, 54pt)
let text-styles = ((weight: 700), (size: 10pt), (size: 9pt, weight: 500), (size: 9pt, fill: luma(20%)), )
theoretic.toc(toc-entry: theoretic.toc-entry.with(
  indent: (level) => { indents.at(level - 1) },
  hanging-indent: (level) => { hang-indents.at(level - 1) },
  fmt-prefix: (prefix, level, _s) => {
    set text(..text-styles.at(level - 1), number-width: "tabular")
    prefix
    h(4pt)
  },
  fmt-body: (body, level, _s) => {
    set text(..text-styles.at(level - 1))
    body
  },
  fmt-fill: (level, _s) => {
    if level == 2 {
      set text(..text-styles.at(2))
      box(width: 1fr, align(right, repeat(gap: 9pt, justify: false, [.])))
    }
  },
  fmt-page: (page, level, _s) => {
    set text(..text-styles.at(level - 1), number-width: "tabular")
    box(width: 18pt, align(right, [#page]))
  },
  above: (level) => {
    if level == 1 {
      auto // paragraph spacing
    } else {
      7pt
    }
  }
)
```

```

    }
  },
  below: auto,
))

```

## Parameters

```

toc-entry(
  <level>: int,
  <target>: location,
  <prefix>: content none,
  <body>: content,
  <page>: content,
  secondary: boolean,
  indent: relative length function,
  hanging-indent: relative length function auto,
  above: relative length function,
  below: relative length function,
  fmt-prefix: function,
  fmt-body: function,
  fmt-fill: function,
  fmt-page: function
)

```

**secondary:** `false` `boolean`

This is true for entries where the toc-title is an array, the entry was split and this is *not* the first one (in order specified).

**indent:** `1em` `relative length` or `function` How much to indent each entry.

- If `relative length`, it will be multiplied with level - 1.
- If `function`, will be called with the level as argument.

**hanging-indent:** `auto` `relative length` or `function` or `auto`

How much more to indent subsequent lines (in addition th `toc-entry.indent`).

If the prefix is shorter than this, this will lead to a gap between prefix and body; If the prefix is longer, the body will start immediately after the prefix.

- If `function`, will be called with the level as argument.
- If `auto`, will use the width of the prefix

```

#let example-entry = theoretic.toc-entry.with(1, here(),
[Section 1.], lorem(6), [0])
#let example-entry-2 = theoretic.toc-entry.with(2, here(),
[Section 1.1.], lorem(6), [0])

// aligned with end of prefix
#example-entry(hanging-indent: auto)
#example-entry-2(hanging-indent: auto)

#example-entry(hanging-indent: 1em)
#example-entry-2(hanging-indent: 1em)
#example-entry(hanging-indent: 80pt)
#example-entry-2(hanging-indent: 80pt)

```

```

Section 1. Lorem ipsum dolor sit amet, consectetur. . . . . 0
  Section 1.1. Lorem ipsum dolor sit amet, consectetur. . 0
Section 1. Lorem ipsum dolor sit amet, consectetur. . . . . 0
  Section 1.1. Lorem ipsum dolor sit amet, consectetur. . 0
Section 1.      Lorem ipsum dolor sit amet, consecte-
                tur. . . . . 0
  Section 1.1.   Lorem ipsum dolor sit amet, con-
                sectetur. . . . . 0

```

**above:** `0.7em` `relative length` or `function` If `function`, will be called with the level as argument.

**below:** `0.7em` `relative length` or `function` If `function`, will be called with the level as argument.

### A.13 thm-counter `counter`

Counts theorems.

In most cases, it is not necessary to reset this manually, it will get updated accordingly if you pass an int to `theorem.number`.

### A.14 proof `function`

This is simply `theorem()` with different options.

## B Solutions

**Solution** (Exercise 2.42). *Yay! you found the solutions!*

**Solution** (Theorem A.16). *This will show up wherever `#theoretic.solutions()` is placed.*